Chapter 8 Assignment — Polar Coordinates and Parametric Equations

Name: ____________________ Date: ____________________

How to use this set: Each topic begins with *Strategy Notes* and a *Worked Example.* Then try the *Practice* (skill) and the *Challenge* (deeper) items. Give exact answers using radicals and π when possible. For polar angles, state units and choose $0 \leq \theta < 2\pi$ unless otherwise specified.

---

1. **Topic 1: Polar Coordinates and Conversions**

   Strategy Notes. Use $x = r\cos\theta$, $y = r\sin\theta$, $r = \sqrt{x^2 + y^2}$, and $\tan\theta = \dfrac{y}{x}$ with quadrant checks. Remember that $(r, \theta) = (-r, \theta + \pi)$ gives the same point.

   Worked Example. Convert the rectangular point $(-5, 0)$ to polar form.

   Solution. $r = \sqrt{(-5)^2 + 0^2} = 5$. Since the point is on the negative x-axis, $\theta = \pi$. One representation is $(5, \pi)$. An alternative is $(-5, 0)$.

   Practice.

   (a) Convert $(6\sqrt{2}, 6)$ to polar coordinates with $r > 0$ and $0 \leq \theta < 2\pi$.

(b) Convert the polar point $(9, -180°)$ to rectangular coordinates.

Challenge.

(a) The rectangular point $(-6, 2\sqrt{3})$ has many polar representations. Find one with $r > 0$ and one with $r < 0$, both with $0 \leq \theta < 2\pi$.

2. **Topic 2: Graphs of Polar Equations**

   Strategy Notes. Recognize families: $r = a\cos\theta$ or $r = a\sin\theta$ are circles; $r = a \pm b\cos\theta$ or $a \pm b\sin\theta$ are limaçons (inner loop if $|a| < |b|$); $r = \sin(k\theta)$ or $r = \cos(k\theta)$ are rose curves (if k is even, $2k$ petals; if k is odd, k petals). Converting to rectangular can reveal the type quickly.

   Worked Example. Graph $r = 8\cos\theta$ and classify the curve.

   Solution. Multiply by r: $r^2 = 8r\cos\theta \Rightarrow x^2 + y^2 = 8x$, so $(x-4)^2 + y^2 = 4^2$. This is a circle of radius 4 centered at $(4, 0)$.

   Practice.

   (a) Graph $r = 3 + 6\sin\theta$. Identify the curve type and indicate intercepts with the coordinate axes.

   (b) Graph $r = \dfrac{4}{1-\cos\theta}$. Convert to rectangular form, identify the conic, and mark key features (vertex/focus/directrix or asymptotes as appropriate).

Challenge.

(a) What does the graph of $r = 2 + 3\theta$ look like for $\theta \geq 0$? Sketch the first two turns and indicate the direction as θ increases.

3. **Topic 3: Complex Numbers in Polar Form and De Moivre's Theorem**

   Strategy Notes. For $z = a + bi$, write $z = r(\cos\theta + i\sin\theta)$ where $r = |z| = \sqrt{a^2 + b^2}$ and $\theta = \operatorname{Arg}(z)$. Then

   $$z_1 z_2 = r_1 r_2 \left(\cos(\theta_1+\theta_2)+i\sin(\theta_1+\theta_2)\right), \quad \frac{z_1}{z_2} = \frac{r_1}{r_2}\left(\cos(\theta_1-\theta_2)+i\sin(\theta_1-\theta_2)\right), \quad z^n = r^n \left(\cos(n\theta)+i\sin\right.$$

   Worked Example. Let $z = 1 + \sqrt{3}\,i$. Write z in polar form and compute z^9.

   Solution. $r = \sqrt{1+3} = 2$, $\theta = \arctan(\sqrt{3}) = \frac{\pi}{3}$. Thus $z = 2\big(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\big)$. Then $z^9 = 2^9\big(\cos 3\pi + i\sin 3\pi\big) = 512\big(\cos\pi + i\sin\pi\big) = -512$.

   Practice.

   (a) For $z_1 = 4\big(\cos\frac{7\pi}{12} + i\sin\frac{7\pi}{12}\big)$ and $z_2 = 2\big(\cos\frac{5\pi}{12} + i\sin\frac{5\pi}{12}\big)$, compute $z_1 z_2$ and $\dfrac{z_1}{z_2}$ in polar form.

   (b) Express $-1 - i$ in polar form (principal argument in $(-\pi, \pi]$).

Challenge.

(a) Compute all fourth roots of $a = 1 + \sqrt{3}$ and describe where they lie in the complex plane.

4. **Topic 4: Parametric Equations and Eliminating the Parameter**

Strategy Notes. For lines, use $(x,y) = (x_0, y_0) + t(\,v_x, v_y\,)$. For ellipses $\dfrac{(x-h)^2}{a^2} + \dfrac{(y-k)^2}{b^2} = 1$, use $x = h + a\cos t$, $y = k + b\sin t$. Eliminate t by using identities like $\sin^2 t + \cos^2 t = 1$.

Worked Example. The curve $x = 3\sin t + 3$, $y = 2\cos t$ has what rectangular equation?

Solution. $\dfrac{(x-3)^2}{3^2} + \dfrac{y^2}{2^2} = \sin^2 t + \cos^2 t = 1$, an ellipse centered at $(3,0)$ with semi-axes 3 and 2.

Practice.

(a) Find parametric equations for the line of slope 2 through $(3,5)$.

(b) Find parametric equations for $4x^2 + 9y^2 = 144$.

Challenge.

(a) Eliminate the parameter for $x = 3\sin(2t)$, $y = 3\cos(2t)$ and identify the curve; indicate the direction of motion as t increases.

5. **Topic 5: Modeling with Parametric Motion**

   Strategy Notes. Uniform circular motion of period T about center (h, k) and radius R can be written as $x = h + R\cos(\omega t + \phi)$, $y = k + R\sin(\omega t + \phi)$ with angular speed $\omega = \dfrac{2\pi}{T}$; choose phase ϕ to match the starting point and sign of rotation.

   Worked Example. If a particle circles counterclockwise once every 8π seconds with center $(4, 5)$ and starts at $(1, 5)$, then $R = 3$, $\omega = \dfrac{1}{4}$ and one model is $x = 4+3\cos\left(\frac{t}{4} + \pi\right)$, $y = 5 + 3\sin\left(\frac{t}{4} + \pi\right)$.

   Practice.

   (a) A particle starts at $(1, 5)$ and travels at a constant speed counterclockwise in a circle around $(4, 5)$, taking 8π seconds to complete one revolution. Give parametric equations for its position t seconds after it starts.

   (b) After doubling the speed, give new parametric equations for the motion (same center and radius, same orientation).